



TRAINING PROGRAMME

Enterprise Virtualization: KVM & Proxmox VE

Design, run and protect a highly available virtualization cluster

36 HOURS

FOR System administrators who run or plan a virtualization platform



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Overview

This programme teaches participants to build and run a virtualization platform on open-source software. They start with KVM, QEMU and libvirt on a single host, then deploy Proxmox VE: networking with bridges, VLANs, bonds and SDN, local and shared storage up to hyperconverged Ceph, virtual machines and LXC containers, automation through the API and Ansible, clustering with high availability and live migration, access control, firewalling and backups with Proxmox Backup Server. They also migrate virtual machines from VMware ESXi and compare the platforms on the market. Every chapter ends with a lab, and the programme closes with a final project in which teams deploy and present a three-node cluster.

At a glance

#	PHASE	FOCUS	HOURS
01	KVM & Proxmox VE Foundations	KVM and libvirt, Proxmox VE networking and storage	9 h
02	Virtual Machines & Containers	VMs, templates and cloud-init, LXC containers and automation	6 h
03	Clustering, Operations & Migration	High availability, access control, backups, migration, platform choice	9 h
04	Final Project	A three-node highly available cluster, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
Total			36 h

Key facts

AUDIENCE	System administrators who run or plan a virtualization platform
PREREQUISITES	Linux administration (course: Linux & Bash Scripting, or RHCSA Part 1: Red Hat System Administration I) and basic IP networking. Linux basics are not taught again
DURATION	36 h: 30 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs
LAB ENVIRONMENT	Runs on premises: one lab server per team of two or three (64 GB RAM, nested virtualization), provided by the organiser, hosting as virtual machines a KVM host, three Proxmox VE nodes, a Proxmox Backup Server and a source hypervisor for migration; Windows evaluation images; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can deploy and operate a Proxmox VE cluster with shared storage, high availability, tested backups and migrated workloads, and choose a platform for their organisation

PHASE 01

9 H

KVM & Proxmox VE Foundations

Participants see how a hypervisor works with KVM, QEMU and libvirt on a single host, then install Proxmox VE and build the networking and storage it needs: Linux bridges, VLANs, bonds and SDN, local LVM-thin and ZFS, shared NFS and iSCSI, and hyperconverged Ceph. Ceph is introduced here and deployed in phase 3, once the cluster exists. Validated lab: a Proxmox VE node with a VLAN-aware bridge, ZFS local storage and an NFS share, managed from the web interface and the command line.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Virtualization & KVM	Hypervisor types, KVM and QEMU, libvirt, virsh, virt-install, Cockpit	Lecture + Lab	3 h
1.2	Proxmox VE Architecture	Installation, web interface and command line, nodes, Linux bridges, VLANs and bonds, SDN	Lecture + Lab	3 h
1.3	Storage	Local LVM-thin and ZFS, NFS and iSCSI shared storage, hyperconverged Ceph	Lecture + Lab	3 h

PHASE 02

6 H

Virtual Machines & Containers

Participants create Linux and Windows virtual machines with VirtIO drivers, turn them into cloud-init templates, and use snapshots and clones. They run LXC containers, then automate the platform through its API, pvesh and Ansible. Validated lab: a cloud-init template cloned into three configured virtual machines by an Ansible playbook.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Virtual Machines	Linux and Windows VMs, VirtIO drivers, cloud-init, templates, snapshots, clones	Lecture + Lab	3 h
2.2	Containers & Automation	LXC containers, the Proxmox VE API and pvesh, automation with Ansible	Lecture + Lab	3 h

PHASE 03

9 H

Clustering, Operations & Migration

Participants join nodes into a cluster with quorum, move running machines with live migration and protect them with high availability. They control access, firewall the platform, back up and restore with Proxmox Backup Server, migrate workloads from VMware ESXi and other hypervisors, and compare Proxmox VE with the alternatives now that Red Hat Virtualization has reached end of life. Validated lab: a virtual machine on Ceph storage restarted automatically on another node after a node failure, then restored from backup.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Clustering & High Availability	Corosync cluster, quorum and QDevice, live migration, HA rules (node and resource affinity), fencing	Lecture + Lab	3 h
3.2	Operations & Security	Users, roles and realms (LDAP and Active Directory), permissions, firewall, updates, Proxmox Backup Server, restore	Lecture + Lab	3 h
3.3	Migrating to Proxmox VE	ESXi import wizard, virt-v2v, OVF and OVA import, migration planning	Lecture + Lab	1.5 h
3.4	The Enterprise Landscape	Red Hat Virtualization end of life, oVirt as a community project, OpenShift Virtualization (KubeVirt), choosing a platform	Lecture + Workshop	1.5 h

PHASE 04

6 H

Final Project

Teams of two or three design and deploy a three-node Proxmox VE cluster on their lab server, reusing the phase labs, then present it as they would to an IT manager choosing a platform.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Design & Build	Read the brief, design the network and storage, then build the three-node cluster with Ceph or shared storage, templates and an HA virtual machine	Project	3 h
4.2	Migrate, Back Up & Document	Migrate a virtual machine from another hypervisor, run and restore a backup, write the architecture diagram and runbook	Project	1.5 h
4.3	Demo & Review	Live migration and a node failure during the demo; questions and feedback	Project	1.5 h

FINAL PROJECT

A Highly Available Virtualization Cluster

A three-node Proxmox VE cluster with Ceph or shared storage, templates, a highly available virtual machine that survives a node failure, live migration, a tested backup and restore plan, and a virtual machine migrated from another hypervisor.

STACK Proxmox VE, KVM, QEMU, LXC, Ceph or NFS, ZFS, Corosync, Proxmox Backup Server, cloud-init, Ansible

DELIVERABLE Working cluster, architecture diagram, backup plan and runbook, live demo